



Unlocking Societal Trends in Aadhaar Enrolment and Updates

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###

Abstract

India's Aadhaar ecosystem has quietly crossed a historic inflection point. What began as the world's largest identity issuance program has now evolved into a mature, steady-state governance platform—one whose greatest challenge is no longer coverage, but continuity. This report reimagines Aadhaar not merely as an identity database, but as a national socio-economic sensor, capable of revealing demographic shifts, service stress, migration dynamics, and welfare risk in near real time.

Leveraging over 5 million Aadhaar records across biometric, demographic, and enrollment datasets, this study presents a 360° analytical framework to diagnose systemic pressure, predict future load, and uncover hidden inequities in India's identity infrastructure. The analysis reveals a striking reality: 98% of new Aadhaar activity is now child-driven, while adult enrollment has effectively collapsed—signaling universal adult coverage. This demographic inversion exposes a critical mismatch between static, adult-centric infrastructure and a rapidly evolving lifecycle-management mandate dominated by infant registrations and mandatory biometric updates at ages 5 and 15.

Through advanced temporal analysis, geospatial intelligence, anomaly detection, and machine-learning-based forecasting, the report uncovers where, when, and why Aadhaar systems strain or fail. It identifies urban-industrial overload zones (Pune, Nashik, Thane), delayed-enrollment cultures (Bihar, Meghalaya), digital darkness regions (Ladakh, Nicobars), and welfare risk districts trapped in a cycle of high child dependency and adult out-migration. Seasonal peaks, weekday overloads, and underutilized weekends reveal opportunities for immediate operational optimization—while shock detection demonstrates Aadhaar's ability to sense displacement events, administrative anomalies, and fraud-prone clusters.

Most critically, the report shifts Aadhaar from a reactive administrative system to a predictive governance engine. Forecasts for 2026 show biometric workloads concentrating sharply in a handful of large states and urban districts, enabling proactive deployment of devices, manpower, and mobile infrastructure. By translating enrollment data into forward-looking insights for education, healthcare, welfare, and workforce planning, this study positions Aadhaar as a foundational tool for anticipatory governance.

In essence, this report demonstrates that Aadhaar's greatest value now lies beyond identity—it lies in its power to see the nation change before the nation feels it. This transformation, if harnessed strategically, can redefine how India plans, allocates, and governs in the decade ahead.

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Strategic Roadmap for Next-Gen Aadhaar Governance

1. Problem Statement and Approach

India's Aadhaar ecosystem has transitioned from a phase of mass identity issuance to a "steady-state" platform focused on lifecycle management. The core challenge is no longer achieving universal coverage but maintaining it—specifically for children who must update biometrics at ages 5 and 15. Static infrastructure designed for adults is increasingly inefficient for a landscape now dominated by infant registrations and mandatory updates.

Analytical Approach:

We utilized a multi-stage pipeline to process over 5 million records. By treating Aadhaar data as a National Socio-Economic Sensor, we integrated three datasets to identify:

- **Infrastructure Pressure Points:** Predicting where biometric hardware will fail due to excessive load.
 - **Welfare Risk Zones:** Identifying "poverty traps" where high child dependency meets adult out-migration.
 - **Service Gaps:** Mapping "Digital Darkness" areas where connectivity or distance prevents enrollment.
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2. Datasets Used

This report synthesizes insights from three comprehensive UIDAI data sources:

Biometric Dataset: ~2.86 million rows across 806 districts tracking authentication spikes (March–Dec 2025).

Demographic Dataset: ~2.07 million rows across 1,077 districts focusing on migration and dependency ratios.

Enrollment Dataset: Granular snapshot of new identity creation across 36 States/UTs (Sept 2025).

3.Key Features:

- 1. Seasonal Peak & Biometric Anomaly Detection System**
- 2. AI-Based Future Biometric Load Forecasting (2026)**
- 3. Geospatial Biometric Hotspot & Digital Divide Mapping**
- 4. Top-Pressure District Identification for Biometric Services**
- 5. Demographic-Driven Aadhaar Enrollment–Based School & Infrastructure Planning**
- 6. Intra-State pin code Variability & Aadhaar Enrollment Coverage Gap Detection**
- 7. Temporal Aadhaar Enrollment Pattern Analysis (Monthly & Campaign Insights)**
- 8. Anomaly, Fraud & Shock Event Detection**
- 9. Digital Service Accessibility & Inclusion Mapping**
- 10. Welfare Risk Zone Identification**
- 11. Migration & Workforce Movement Intelligence**
- 12. Adult Change:** A proxy for internal migration.
- 13. Dependency Ratio:** Measuring the economic burden of children on the working population.
- 14. Composite Stress Index:** A metric combining workforce depletion, high child share, and welfare vulnerability.

4. Methodology & Data Processing

To process high-volume demographic and enrollment data efficiently, a structured multi-stage analytical pipeline was implemented.

1. Data Ingestion & Validation

- Multi-file data ingestion from multiple datasets
- Schema validation to ensure column consistency across files.
- Removal of corrupted or invalid geographic records.

2. Data Cleaning & Standardization

- Date parsing and conversion to datetime64[ns] format for time-based analysis.
- Numeric cleaning, missing value handling, and type optimisation (e.g., enrollment counts converted from int64 to int32).
- Geographic normalisation of State and District names (case standardisation) to eliminate duplication.

3. Data Optimisation

Conversion of CSV files to Parquet format to reduce storage size and improve processing speed.

4. Data Transformation & Aggregation

- Targeted temporal filtering to isolate September data for snapshot analysis.
- Demographic filtering to focus on the 0–5 age group (Bal Aadhaar).
- Aggregation performed at Pincode → District → State levels.

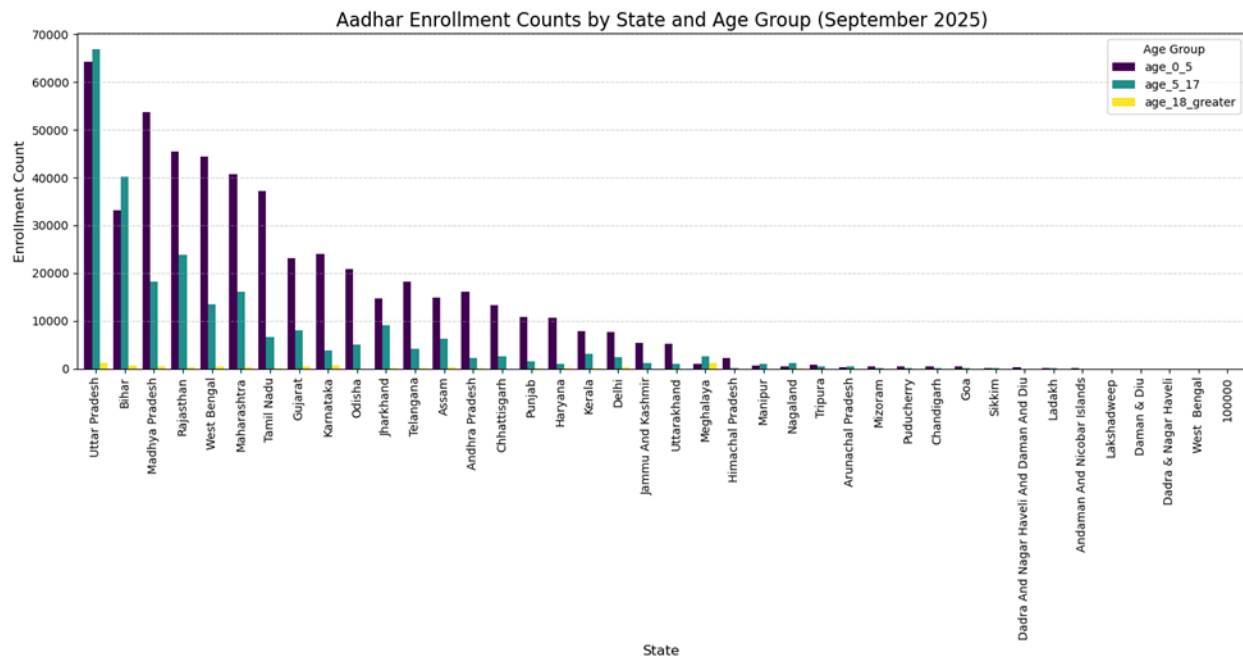
5 .Advanced Analytics & Visualization:

- Multiple visual techniques—including choropleth maps, heat maps, box plots, bar and clustered bar graphs, and line graphs—were used to analyse spatial, temporal, and demographic enrollment patterns.

These visualisations enabled clear identification of high-intensity regions, anomalies, and trends, simplifying interpretation of complex data.

5. Comprehensive Data Analysis and Insights

SECTION 1: System Monitoring & Baseline Understanding



Mapping India's Infant vs School-Age Enrollment Divide

Adult enrollment (18+) has collapsed to negligible levels (making up only 0.9% of activity in states like Uttar Pradesh), confirming universal coverage. New enrollment is now 98% child-driven, split into two distinct zones:

Zone A: In these states, Infant Enrollment (0-5) is massively higher than School-Age Enrollment (5-17).

- *MP Ratio:* ~3:1 (53k vs 18k)
- *West Bengal Ratio:* ~3.3:1 (44k vs 13k)
- *Tamil Nadu Ratio:* ~5.6:1 (37k vs 6k)

These states have highly effective birth registration integration. The demand is at the Maternity Ward. Deploy "Tablet-based Discharge Desk" operators in government hospitals.

Zone B: school-age enrollment exceeds infant enrollment.

- *Bihar*: 40k (5-17) > 33k (0-5).
- *Meghalaya*: 2.5k (5-17) > 1k (0-5) (A massive 2.5x gap).

This reveals a "Delayed Entry" culture where . Children are not getting enrolled at birth but are being forced to enroll later due to school admission mandates (e.g., Mid-Day Meal requirements).

Meghalaya Anomaly: (North-East Specific Insight)

- **Observation:** Meghalaya is a unique outlier. It has very low 0-5 enrollment (1,077) but a sudden spike in 5-17 enrollment (2,593).
- **Inference:** This suggests a cultural or administrative barrier to early enrollment in tribal belts. Parents do not see the "Utility" of Aadhaar for a baby. The demand is purely "Compliance Driven" (for school).
- **Impact:** This creates a 5-Year Data Gap where the child is invisible to the state.
- **Solution:** A targeted "Anganwadi Awareness Campaign" is needed specifically for the Seven Sisters states to explain the benefits of Bal Aadhaar *before* school age.

3. Fraud / Systemic Irregularity Detection

Districts with repeated anomaly clusters:

- Tirunelveli (TN)
- Hyderabad (TG/AP)
- Kottayam (Kerala)
- Gariyaband (Chhattisgarh)

Possible causes:

- Institutional enrollment drives
- Labour camps
- Administrative bulk updates
- Synthetic identity risk

These regions are ideal candidates for targeted audits.

4.National Aadhaar Activity Trend

The time series shows:

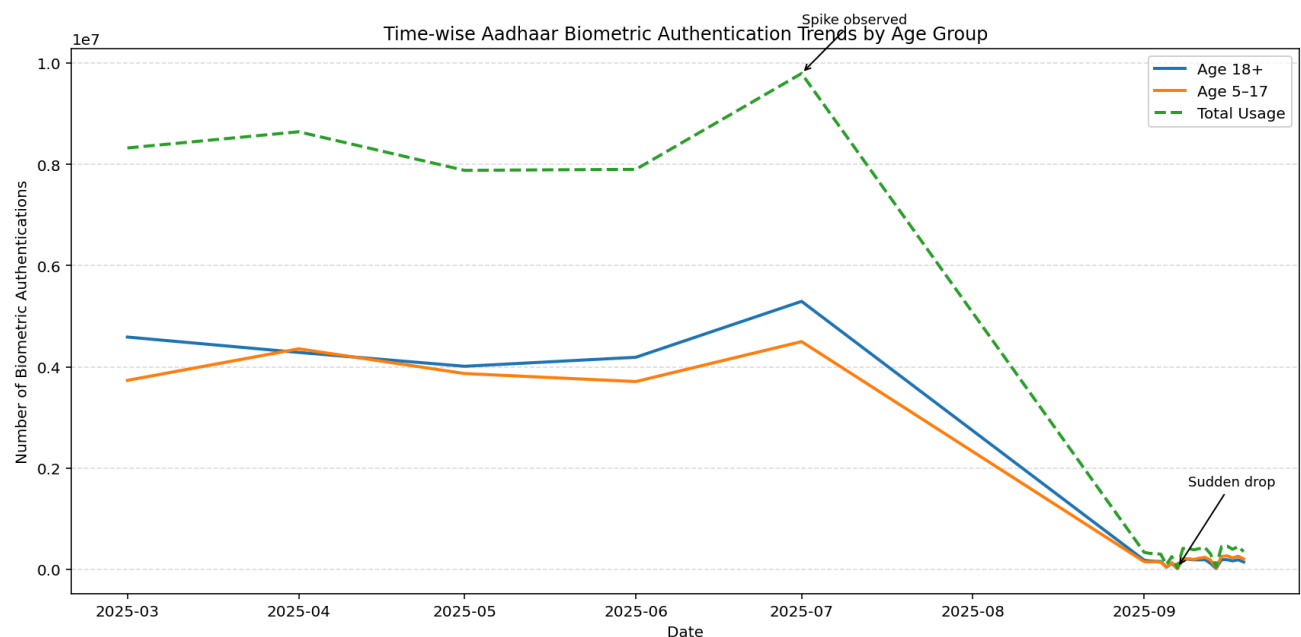
- One early spike (initial mass enrollment phase)
- Stable long-term transaction volume afterwards

Interpretation:

India's identity system has matured into a steady-state operational platform, suitable for long-term demographic sensing and planning.

SECTION 2: Temporal & Operational Intelligence

1.Daily Age Group Wise Biometric Enrollment Trend



Analysis of authentication trends reveals significant fluctuations throughout 2025, including observed spikes in July and sudden drops in September.

- Adult Dominance: Adults (18+) consistently contribute approximately 50.8% of all biometric updates. This indicates that Aadhaar is deeply integrated into daily life

for banking, jobs, and welfare, often overriding policy-driven child update mandates.

- **Backlog Clearance:** High adult volume suggests older enrollees are addressing outdated data from the initial Aadhaar rollout to maintain service access.
- **Temporal Peaks:** Activity peaks in the early part of the year and dips mid-year, likely aligned with financial year beginnings and academic cycles.

Month_Year	Adult Contribution (%)
2025-03	55.14
2025-04	49.58
2025-05	50.91
2025-06	53.03
2025-07	54.06
2025-09	45.75
2025-10	51.66
2025-11	50.46
2025-12	46.88
Average	50.80

2. Temporal Trends: Daily Enrollment Patterns Over the Month

Key Findings and Insights:

- A **significant mid-year spike (June–July)** across all age groups indicates that enrollments are **campaign-driven**, likely aligned with **school admissions and government welfare initiatives**.
- Enrollment activity drops sharply after peak periods, confirming that demand is **episodic rather than continuous**.

Why This Analysis Matters:

- Identifies **peak demand periods**, enabling better planning instead of reactive management.
- Highlights **which age groups drive enrollment**, allowing targeted policy interventions.
- Helps prevent operational overload during high-demand months and underutilisation during low-demand periods.

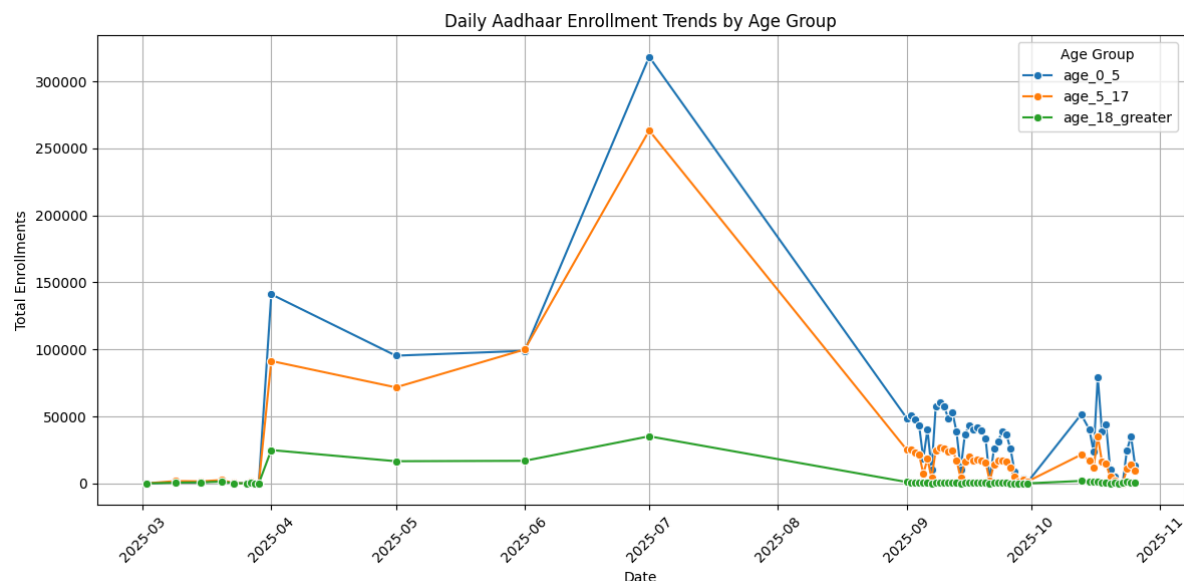
How This Analysis Helps UIDAI and MSMEs:

- **UIDAI** can optimize **resource allocation, manpower deployment, and campaign timing** based on predictable enrollment patterns.
- **MSMEs** supporting enrollment infrastructure can **forecast demand, plan inventory and workforce**, and develop **targeted solutions** for high-impact periods.
- Overall, the analysis supports **data-driven decision-making, cost efficiency, and improved service delivery**.

3. Day-of-Week Enrollment Comparison

Key Findings and Insights:

- Average Aadhaar enrollments are **highest on weekdays**, especially **Tuesday to Thursday** for the **5–17 age group**, likely due to school-based integration.
- **Weekend and Sunday enrollments are 50–70% lower** across all age groups.
- The **0–5 age group shows stable enrollment** throughout the week, indicating hospital-linked registrations, while **18+ enrollments remain minimal**, with a slight mid-week increase.
- This pattern reveals **underutilised weekends** as an opportunity for targeted enrollment initiatives.

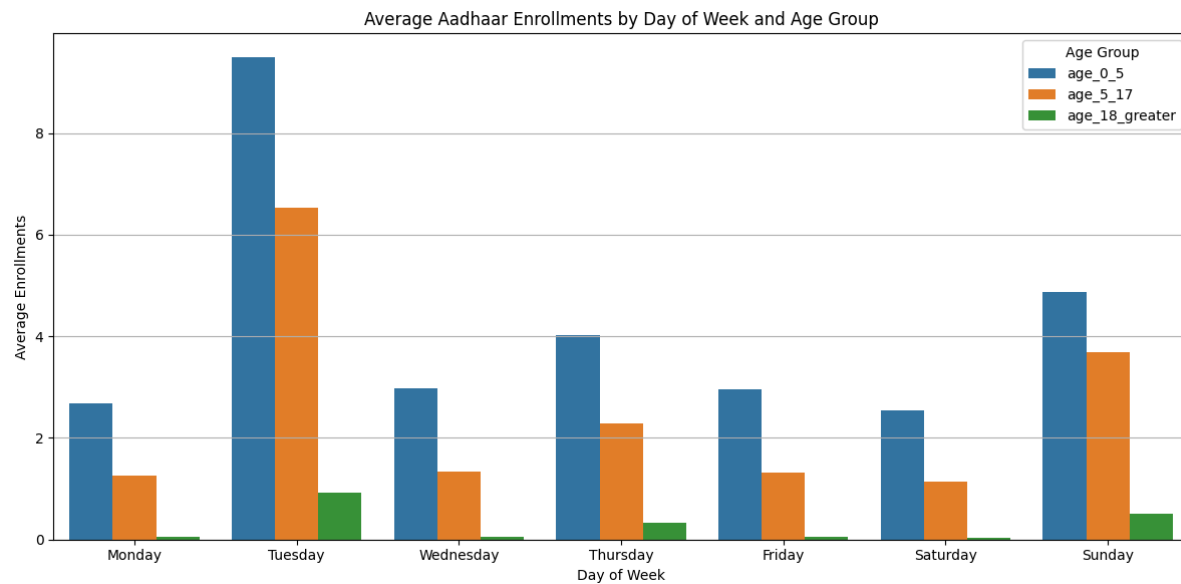


Why This Analysis Matters:

- Understanding day-wise enrollment patterns helps identify **when enrollment centres are overused or underused**.
- This enables better scheduling such as extending hours on high-demand weekdays and introducing weekend incentives **improving access for working parents** and increasing overall enrollments **without adding new infrastructure or major costs**.

How This Analysis Will Help UIDAI or MSMEs:

- For **UIDAI**, the analysis supports **strategic policy planning**, such as introducing weekend drives, mobile camps, or school-hospital partnerships to balance enrollment loads.
- For **MSMEs**, it provides **predictable demand insights**, allowing them to plan staffing, device usage, and service.
- This will offer more efficiency, leading to **better resource utilisation and improved operational returns**



4. Disaster & Shock Detection

Sharp negative population shocks detected in: North Delhi, Central Delhi

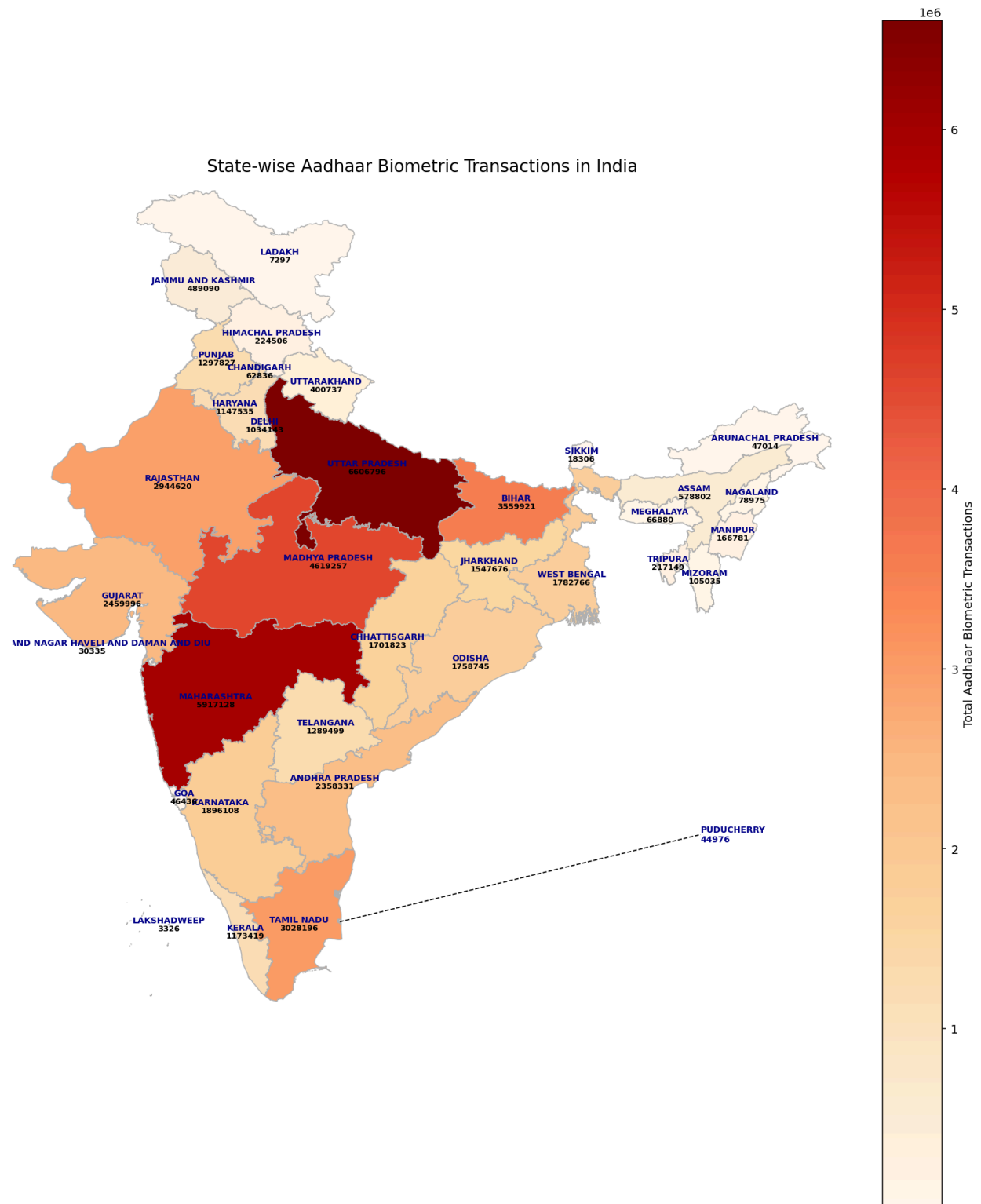
Large monthly drops exceeding 2,000 adults indicate sudden displacement events such as:

- COVID lockdowns
- Demolitions
- Relocations
- Natural disasters

Aadhaar updates provide near-real-time displacement monitoring capability.

SECTION 3: Spatial Equity & Access Intelligence

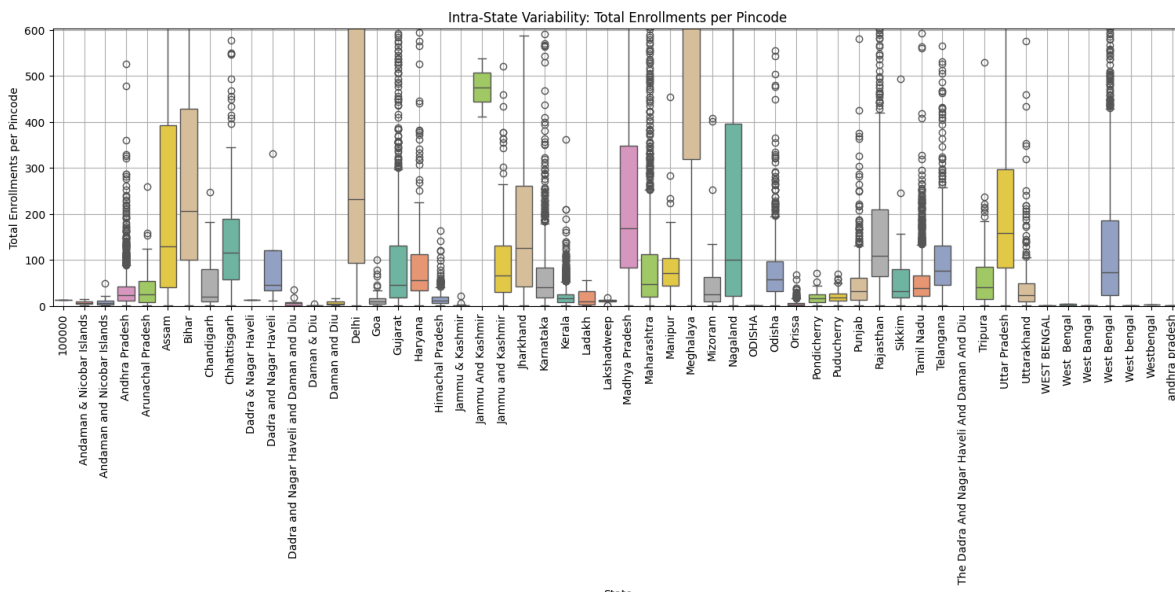
1.State-Wise Aadhaar Biometric Updates Analysis



3. Intra-State and Pincode Variability

Key Findings and Insights:

- The boxplots show how enrolments vary across pincodes within each state.
- States like Uttar Pradesh show high variation, meaning enrolment is uneven, while Kerala shows more uniform coverage.



Why This Analysis Matters:

- This analysis helps uncover hidden gaps within states that state-level averages miss.
- It highlights exactly where enrolment access is inconsistent, allowing authorities to address local imbalances and improve fairness in service delivery.

How This Analysis Helps UIDAI and MSMEs:

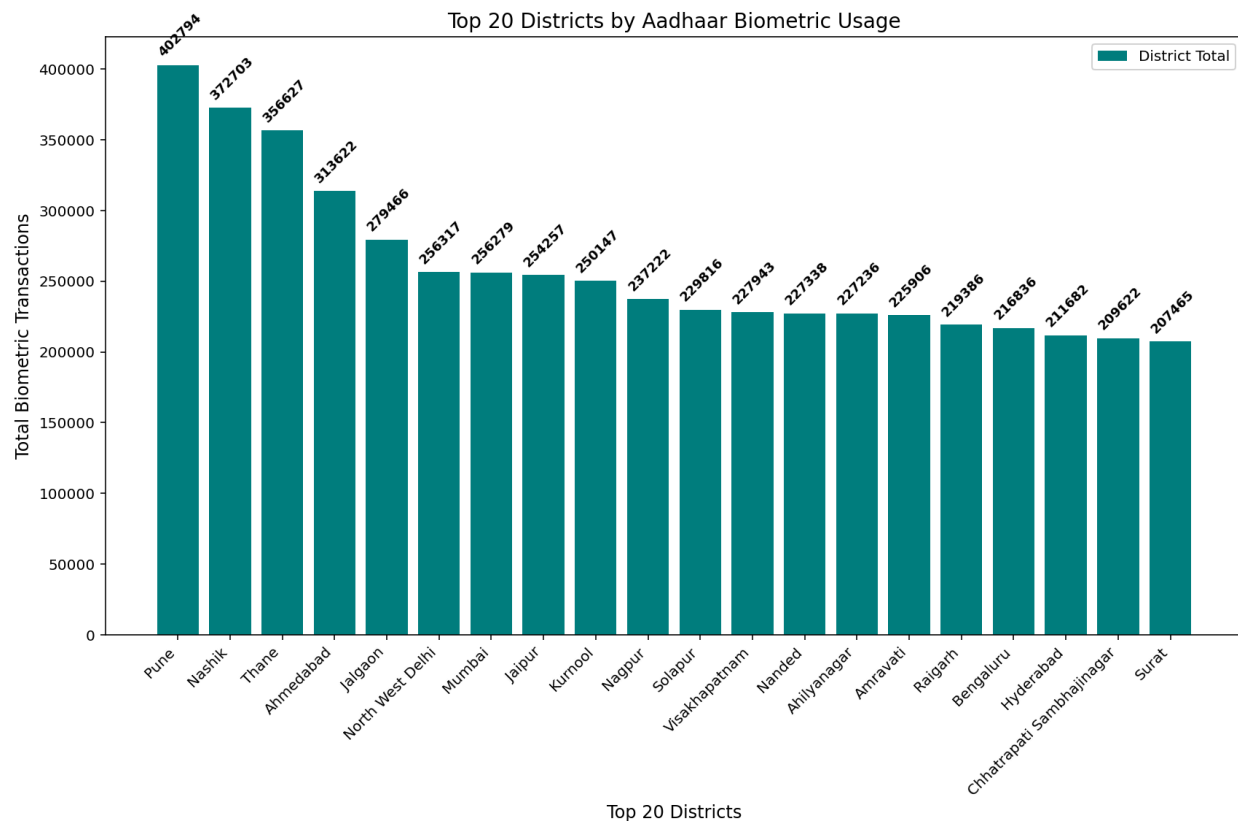
- UIDAI can use this insight to plan pincode-level interventions instead of broad state-wide actions.
- MSMEs and enrolment agencies benefit by knowing where to place mobile units, assign staff, and schedule camps more effectively, leading to smoother operations and better coverage.

- The segmentation of pincodes into "Zones" as **Resource Optimization Strategy would be beneficial**. In logistics and public planning, this is often called "Service Tiers."

Biometric activity shows a strong positive correlation with population density across Indian states.

- High-Volume Leaders: Uttar Pradesh (8.6M+ transactions), Maharashtra (5.9M+), and Madhya Pradesh (4.6M+) dominate national totals.
- Mid-Range States: West Bengal (1.7M+), Karnataka (1.8M+), and Rajasthan (2.9M+) show moderate activity.
- Access Barriers: Low counts in smaller states (Ladakh, Goa) and the North-East (Arunachal Pradesh, Sikkim) highlight significant access barriers despite mandates.
- Inference: Large states account for the vast majority of national activity. While this allows for economies of scale, it risks overwhelming infrastructure, necessitating proactive resource deployment.

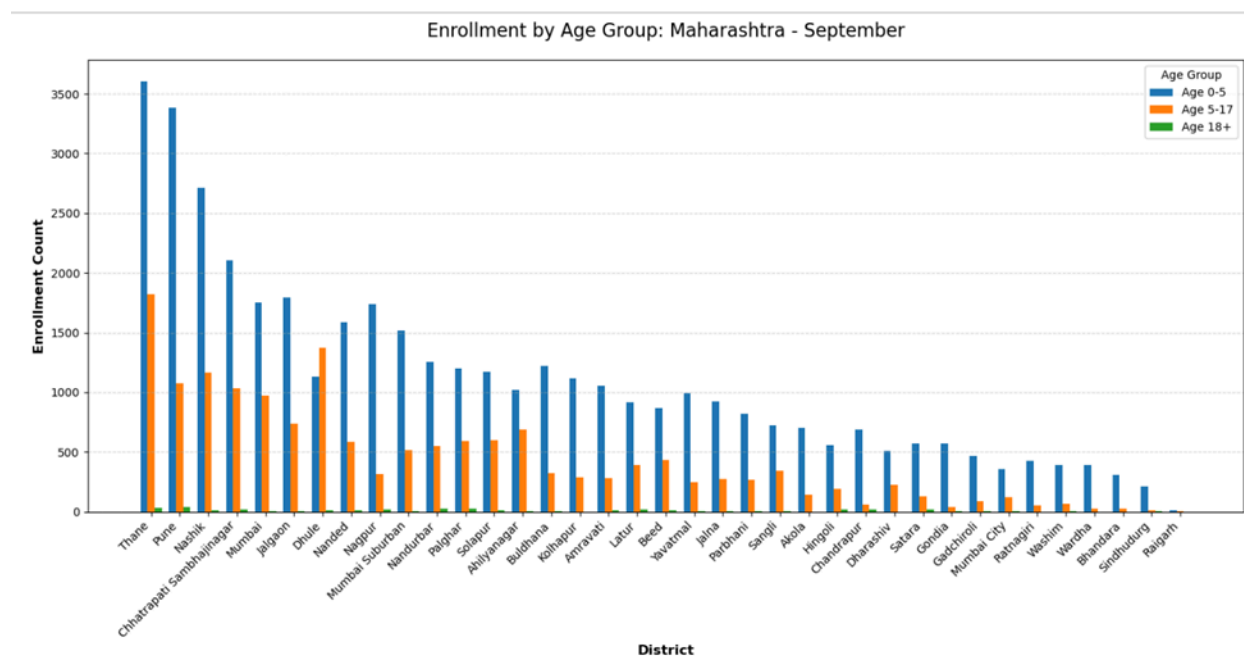
2. Top 20 Districts in Biometric Update in Maharashtra



The "Top 20" districts act as the primary engines of the Aadhaar ecosystem.

- **Urban-Industrial Clusters:** Pune (402k+), Nashik (372k+), Thane (356k+), and Ahmedabad (313k+) lead the rankings. These districts host large industrial labor forces that require biometric refreshes for employment and banking.
- **Infrastructure Edge:** The high volume in these districts is supported by a high density of Aadhaar Seva Kendras (ASKs) and proactive enrollment drives.
- **Digital Divide:** Disparities between these urban hubs and rural districts reveal a digital divide, where urban-rural splits affect welfare reach.

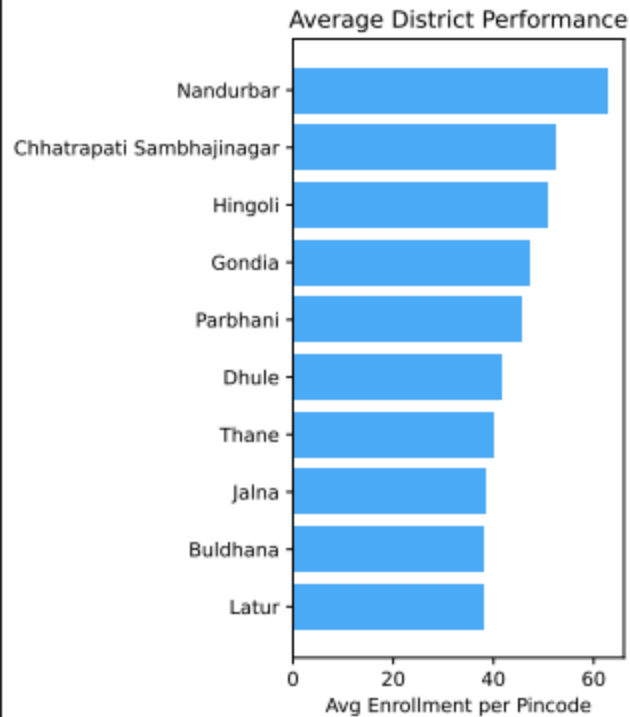
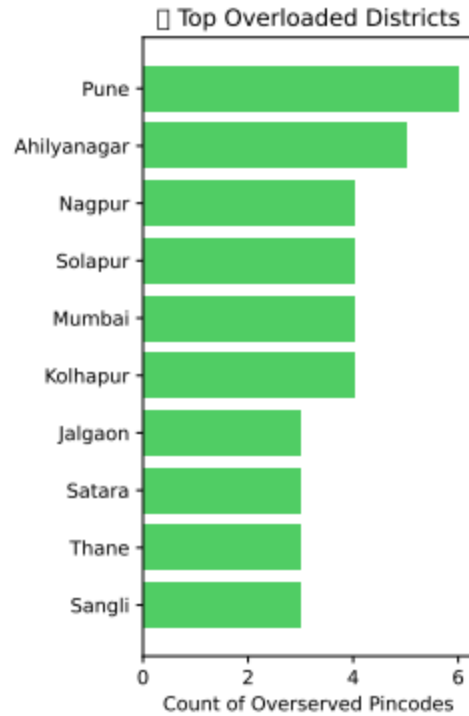
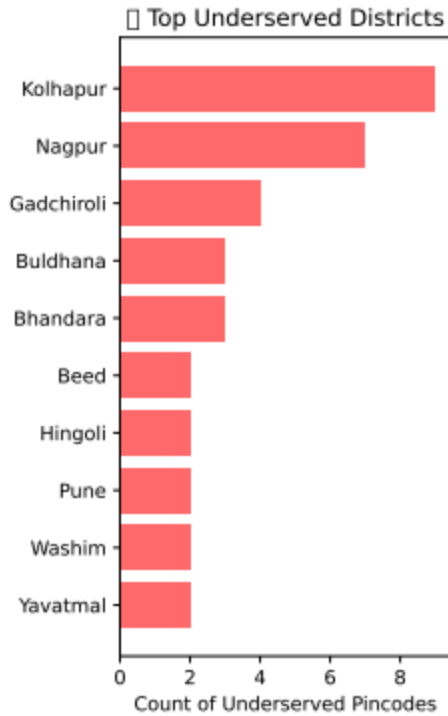
District analysis for Maharashtra



A. The Urban Magnets

- Thane, Pune, Nashik these 3 districts alone account for ~25% of the state's total infant enrollment. Pune ranks as the #1 Overloaded District.
- Root Cause: "Medical Migration." Families from rural districts travel to these healthcare hubs for maternity services, artificially inflating urban performance while masking rural deficits.
- Analysis of Disparity: Pune exhibits a unique "Fringe Latency." While its core is overloaded, it simultaneously ranks #7 in underserved pincodes, indicating infrastructure has failed to keep pace with peri-urban expansion (IT corridors/new suburbs).

State-Level Analysis: September



OVERALL STATISTICS

Total Districts:	37
Districts with Issues:	37
Avg Enrollment (All):	28
Median Enrollment:	28
Total Underserved Pincodes:	46
Total Overserved Pincodes:	75
Normal Pincodes:	1532
% of Districts with Issues:	100.0%

- Kolhapur & Nagpur These districts appear in the Top 5 for BOTH Underserved (Red) and Overloaded (Green) lists.
- Kolhapur: #1 Underserved, #5 Overloaded.
- Root Cause: Resource Misallocation. Enrollment kits are sitting idle in low-demand pockets while centers just kilometers away face unmanageable queues.
- Insight: This is an efficiency problem, not a shortage problem
- Nandurbar: Highest Average Enrollment per Pincode (62) in the state, outperforming IT hubs.
- Gadchiroli: Ranks #3 in Underserved areas with near-zero activity.
- Insight: Nandurbar's success proves that "Camp Mode" operations are highly effective in tribal regions. Gadchiroli's failure indicates a reliance on static centers that the population cannot access.

Summary Table: Utility for Decision Making on the basis of enrollment numbers

Feature	Zone 1 (Peak)	Zone 2 (Middle)	Zone 3 (Low)
Primary Goal	Maximize Volume (Enrolling max kids/day)	Balance Cost vs. Reach	Maximize Coverage (Reaching remote areas)
Best Location	Permanent Shop / Gov Office	School Classroom / Hall	Inside a Van / Open Ground
Duration	365 Days / Year	1-2 Days per Month	2 Hours per Month

Staffing	Dedicated Team (Static)	Roving Team (Camp Mode)	Roving Team (Van Mode)
Kit Type	Desktop / Heavy Duty	Laptop / Portable Kit	Tablet / Ultra-Portable

3. Migration Structure of India (District Level)

Migration category distribution across 1,077 districts:

Category	Districts	Share
High Out-Migration	241	22.4%
Moderate Out-Migration	453	42.1%
Stable	299	27.8%
Moderate In-Migration	56	5.2%
High In-Migration	28	2.1%

Key insight:

Nearly **64% of Indian districts are net population exporters**, while only **7% act as major absorbers**. This confirms a highly concentrated labour market structure with strong regional inequality.

4. Major Migration Destinations (Smart City Candidates)

Top in-migration districts:

- North East Delhi (Delhi)
- Bishnupur (Manipur)
- Kawardha (Chhattisgarh)
- Thoubal (Manipur)
- Uttar Dinajpur (West Bengal)

- Gondiya (Maharashtra)
- Aurangabad (Maharashtra)

Insight:

Urbanisation in India is increasingly decentralised. Growth hubs are not limited to metros, but include tier-2 and tier-3 districts driven by manufacturing clusters, infrastructure projects, and regional service economies.

5. Major Migration Source Regions

Districts with sustained adult population loss:

- Chandauli (Uttar Pradesh)
- Kangpokpi (Manipur)
- Leparda (Arunachal Pradesh)
- Budgam (Jammu & Kashmir)
- Dantewada (Chhattisgarh)

These districts act as long-term workforce exporters.

6.Digital Service Accessibility Gaps

Districts with persistently low Aadhaar activity:

- Leh (Ladakh)
- Nicobars (A&N Islands)
- South 24 Parganas (WB)
- Rupnagar (Punjab)

Meaning:

Limited enrollment infrastructure, remote geography, or digital exclusion. These areas benefit from mobile enrollment vans and service expansion.

SECTION 4: Socio-Economic Intelligence & Governance Insights

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- **Inference:** Large states account for the vast majority of national activity.

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- Digital Divide: Disparities between these urban hubs and rural districts reveal a digital divide, where urban-rural splits affect welfare reach

3. School_Infrastructure_Plan

Key Findings & Insights:

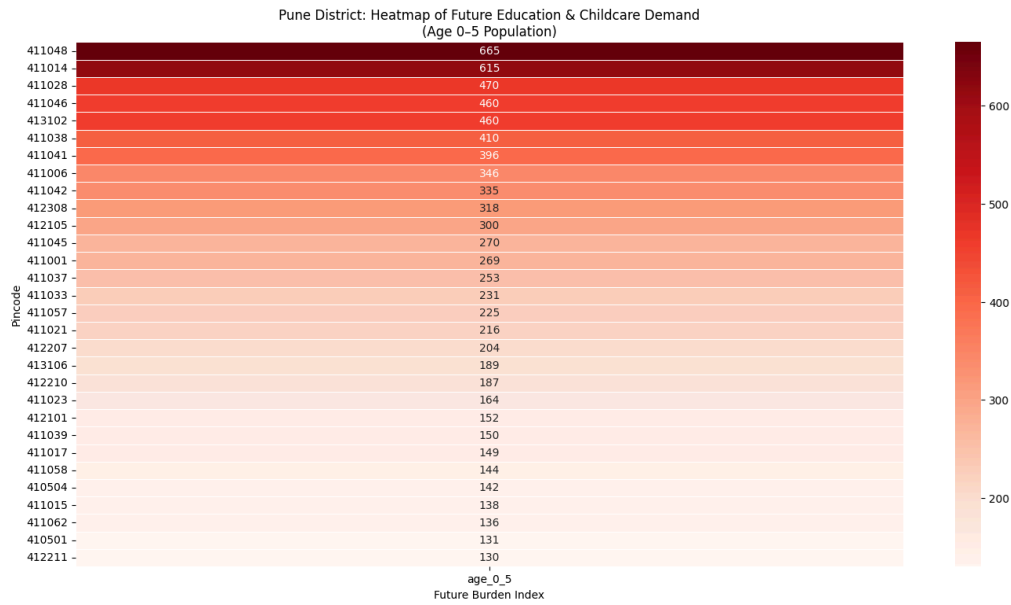
- The heatmap shows clusters of high **age_0_5** populations in a particular district's pincodes, indicating uneven demographic growth.
- High-intensity zones (red) highlight areas where Anganwadis and early-education infrastructure may soon be overburdened.
- These concentrations act as early signals for future demand:
 - **Primary schools** in ~5 years
 - **Secondary schools** in ~10 years
- Transforming enrollment data into forward-looking insights allows authorities to plan resources proactively rather than reactively.

Why This Analysis Matters:

- **Prevention of Overcrowding:** Early identification reduces risks of overcrowded Anganwadis, schools, and poor child-to-worker ratios.
- **Direct Administrative Utility:** The output is district-specific, actionable, and immediately usable by the District Collector, Zilla Parishad, and education planners.

How This Analysis Helps UIDAI and MSMEs:

- **UIDAI:** Helps prioritise Aadhaar registration in high-density child areas, optimise staff and resources, and plan future infrastructure proactively.
- **MSMEs:** Highlights areas with growing child populations, guiding market strategy, expansion, and customised child-focused products or services.



4. Child Dependency Burden (Economic Stress Indicator)

Districts with extreme child dependency ratios:

- Kangpokpi (Manipur)
- Leh (Ladakh)
- Chandauli (UP)
- Leparda (Arunachal Pradesh)
- Budgam (Jammu & Kashmir)

Implications:

High future demand for education, healthcare, nutrition, and welfare spending. These districts represent early warning zones for long-term fiscal and social pressure.

5. Welfare Risk Zones (High child share + adult out-migration)

Examples:

- Budgam (J&K)
- Kangpokpi (Manipur)
- Medchal–Malkajgiri (Telangana)
- Baghpat (UP)
- Chandauli (UP)

Interpretation:

These districts are losing working adults while retaining high child populations — a classic poverty-trap signature.

Such regions require targeted interventions:

- Nutrition programs
- School subsidies
- Direct benefit transfers
- Healthcare expansion

6. Workforce Supply Reservoirs

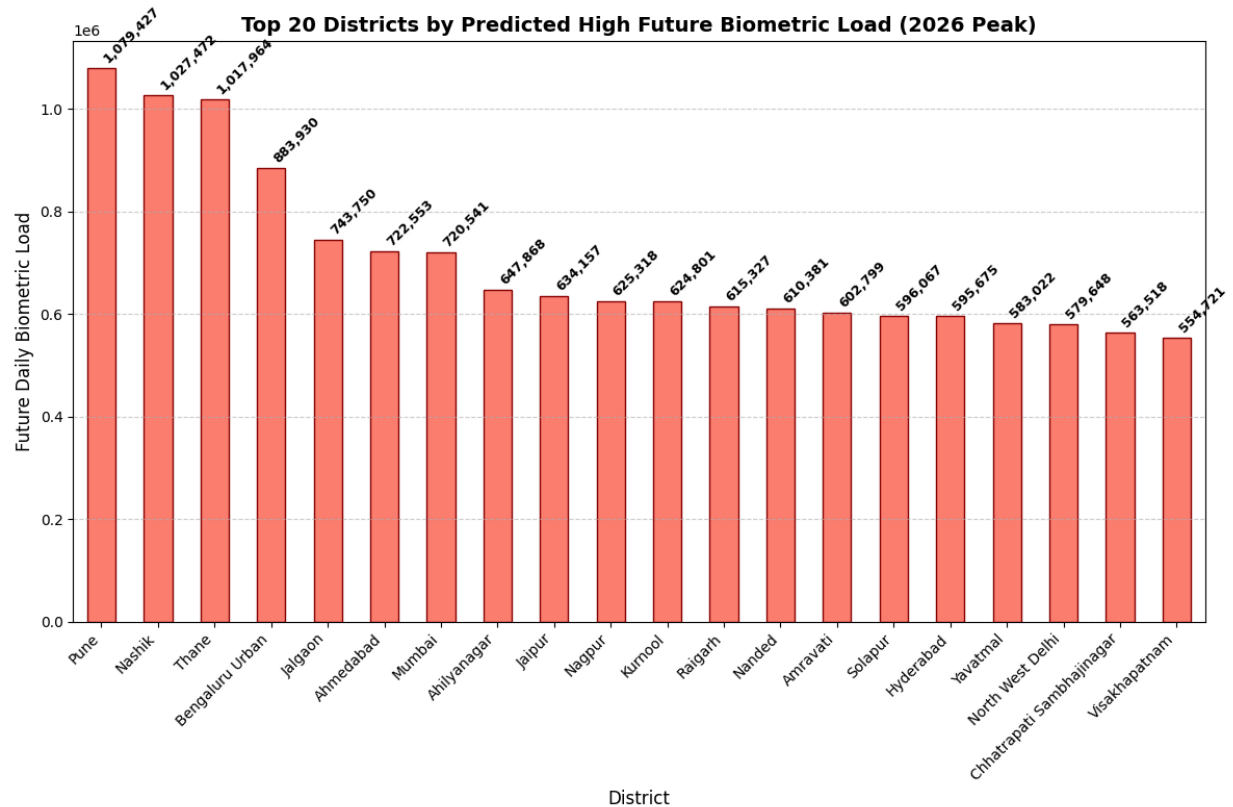
States dominating adult Aadhaar activity:

State	Adult Activity Proxy
Uttar Pradesh	7.75M
Maharashtra	4.78M
Bihar	4.44M
West Bengal	3.63M
Rajasthan	2.56M

These states form India's primary labour supply base.

SECTION 5: Predictive & Future Readiness Layer

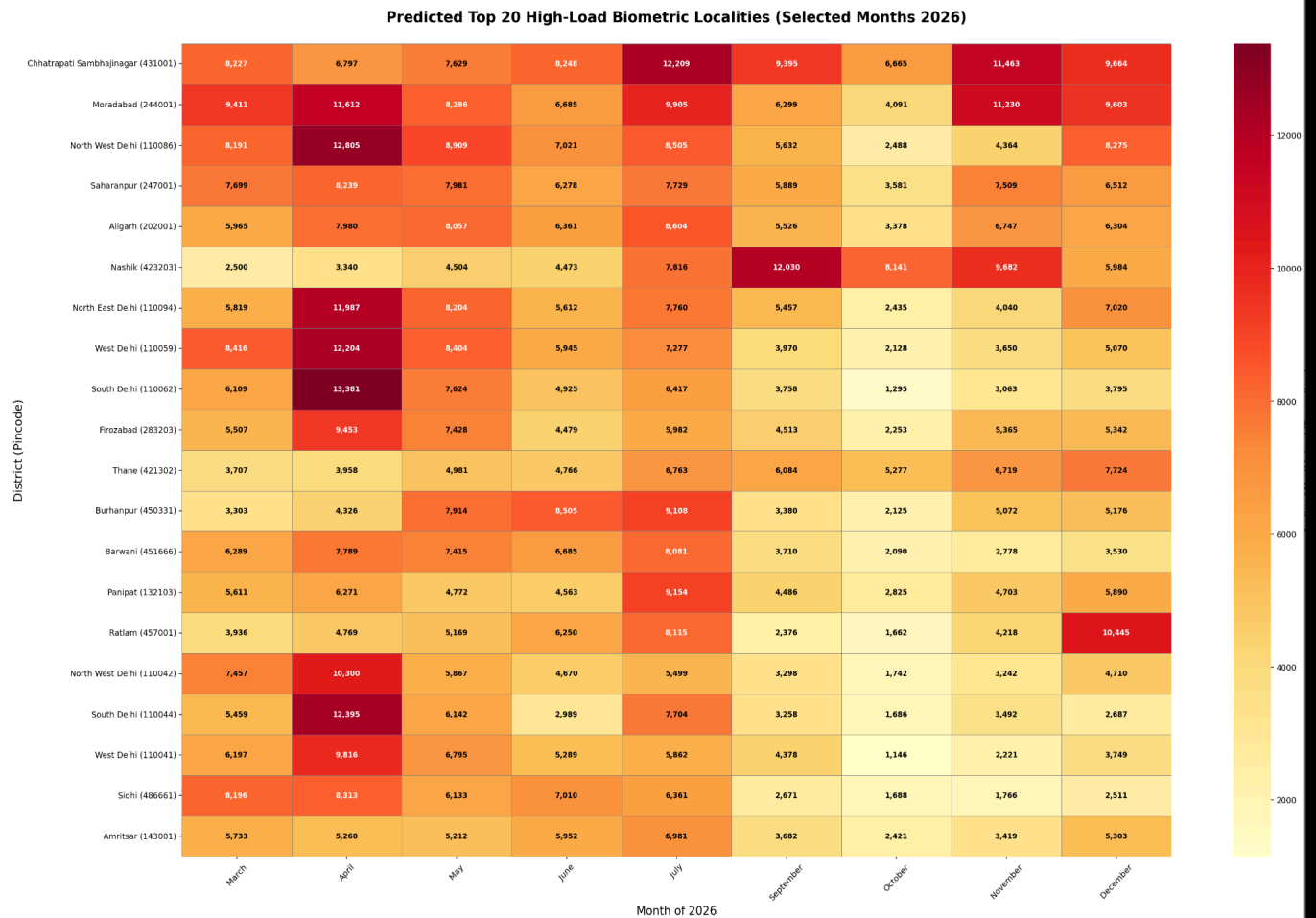
1. Predicted District Load for 2026 Peak



Using the forecasting model, we anticipate continued pressure on major urban centers.

- **Top Forecasted Districts:** Pune, Nashik, and Thane are predicted to maintain the highest daily biometric loads, exceeding 1 million transactions at peak periods in 2026.
- **Rising Hubs:** Bengaluru Urban, Ahmedabad, and Jaipur are also expected to see significant surges, indicating a need for expanded infrastructure.

2. Monthly Seasonal Heatmap of Top Spots with high possibility of load



The granular heatmap identifies specific months where transaction volumes will exceed system thresholds.

- **Equipment Readiness:** Localities like South Delhi, North West Delhi, and West Delhi show intense "high-load" zones specifically during April.
- **Operational Requirement:** These peak periods require a proper supply of devices and additional manpower to prevent system outages.
- **Note on Anomalies:** Occasional low counts in historical data may be attributed to localized system outages or administrative pauses, which must be factored into future capacity planning.

3.National Biometric Update Workload Forecast

States with highest future child biometric update demand:

State	Children Updates Proxy
Uttar Pradesh	790,308
Madhya Pradesh	407,098
Bihar	380,023
Andhra Pradesh	321,143
Tamil Nadu	315,638
Maharashtra	273,322
Karnataka	264,981

Insight:

Over 60% of future biometric workload is concentrated in a small group of large states. This enables UIDAI to plan enrollment kits, operator deployment, and budgets spatially and proactively.

6. Conclusion

- Aadhaar has achieved near-universal adult coverage, with adult enrollment now negligible; the system is fully child-driven (~98%).
- Southern states (e.g., Tamil Nadu) show strong early infant registration (~5.6:1), while Bihar and Meghalaya exhibit delayed school-driven enrollment.
- Enrollment activity is spatially uneven: urban-industrial districts (e.g., Pune, Nashik, Thane) dominate, while remote or smaller regions face access gaps, highlighting digital and infrastructure divides.
- Temporal patterns reveal campaign-driven peaks mid-year, weekday dominance, and underutilized weekends, suggesting opportunities for resource optimization.
- Anomaly and shock detection show that the system can identify bulk

administrative updates, fraud risk zones, and sudden population displacements (e.g., >2,000 adults/month in Delhi districts).

- Migration and socio-economic insights: majority of districts are net population exporters, creating high child-dependency regions that require targeted welfare, education, and healthcare planning.
- In 2026, high biometric workloads will concentrate in large states (UP, MP, Bihar) and major urban districts, enabling proactive allocation of staff, devices, and infrastructure to prevent bottlenecks.
- Overall, Aadhaar functions not only as a registration system, but also as a demographic monitoring, governance, and strategic planning tool, enabling targeted interventions and operational efficiency.

7. Technical Implementation and Accuracy

- **Programming & Libraries:** Python (Pandas, NumPy, Matplotlib, Scikit-learn, BaseGeometry, Prophet) with OS utilities for workflow automation.
- **Architecture:** Modular pipeline enabling production-grade data cleaning, feature engineering, and reproducible dataset and visualization generation.
- **Machine Learning & Anomaly Detection:** Isolation Forest detects systemic fraud and bulk-update clusters, flagging zones like Tirunelveli, Hyderabad, and Gariyaband.
- **Analysis & Insights:** Strong correlation between infant and school-age enrollment ($r = 0.8636$); outliers such as Dhule highlight policy intervention targets. Population shocks in North and Central Delhi (>2,000 adults/month) demonstrate monitoring of displacement events.
- **Reporting:** Automated outputs support dashboards, anomaly reporting, and policy evaluation.

Source code of the analysis can be found here :

<https://github.com/ShreyaTheExplorer/UIDAI-data-hackathon-2026.git>

